

BIG DATA CAPSTONE PROJECT

Healthcare Insurance Analytics Platform

An End-to-End Scalable Big Data Solution for Analyzing Scattered Health Data, Personalizing Offers, and Optimizing Business Revenue Streams.

PREPARED BY

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DATE

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KEY SERVICES

AWS · PySpark · Redshift

Agenda & Project Overview

01/14



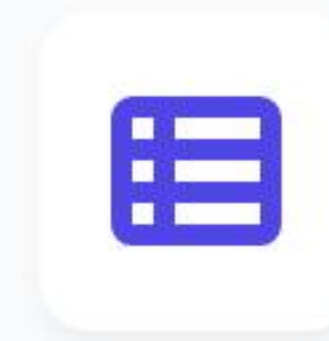
5

Raw Datasets Ingested



13

Business Use Cases Met



18

Tables Loaded in Redshift

01

Problem Definition

Identifying key painpoints & goals

02

Pipeline Design

End-to-end data pipeline architecture

03

Ingestion & Cleaning

AWS S3 layers & PySpark processing

04

Warehouse Load

Redshift loading & analytics scripts

The Challenge & Problem Statement

02/14

"A healthcare insurance provider requires competitive and customer insights to dynamically customize insurance plans, evaluate royalties, and foster growth. Currently, data resides in scattered, unstandardized, and unanalyzed silos."



Raw Scattered Data

5 disparate CSV source tables across distinct locations filled with nulls, exact duplicate entries, and inconsistent data types.



No Analytics Layer

Absence of a consolidated modern cloud data warehouse to query data securely and execute complex analytical workflows at scale.



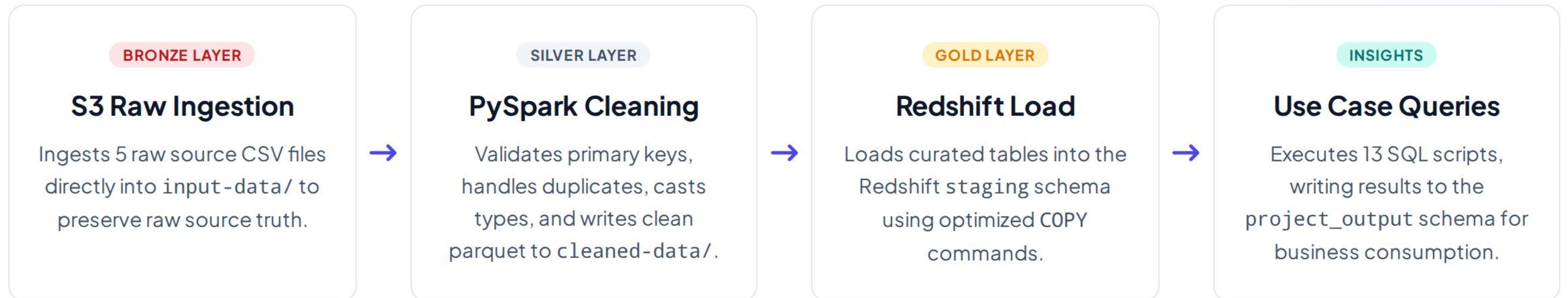
Unanswered Insights

13 fundamental business questions (e.g., profitability, top disease claims, state counts) remained completely unresolvable.

End-to-End Pipeline Architecture

03/14

This multi-stage Big Data pipeline utilizes S3 for storage layers, PySpark for distributed transformations, and Amazon Redshift Serverless for scalable analytics query execution.



AWS S3 Setup: Clean Data Storage Layers

04/14

S3 Bucket Directory Strategy

To decouple compute and storage, we established three strictly segregated layers inside Amazon S3 to organize data progression cleanly.

- ✓ **Bronze Ingestion:** Preserves raw source inputs directly. Immutable folder structure preventing any transactional data loss.
- ✓ **Silver Consolidation:** Converts standard formats into highly optimized Apache Parquet format to improve compression & Spark performance.
- ✓ **Gold Outputs:** Stores highly structured query outputs. Feeds Redshift clusters efficiently.

S3: //HEALTHCARE-CAPSTONE-BUCKET/



input-data/

Stores 5 raw CSV datasets directly from sources



cleaned-data/

Processed Parquet folders (Patients, Subscriber, Claims, etc.)



project-output/

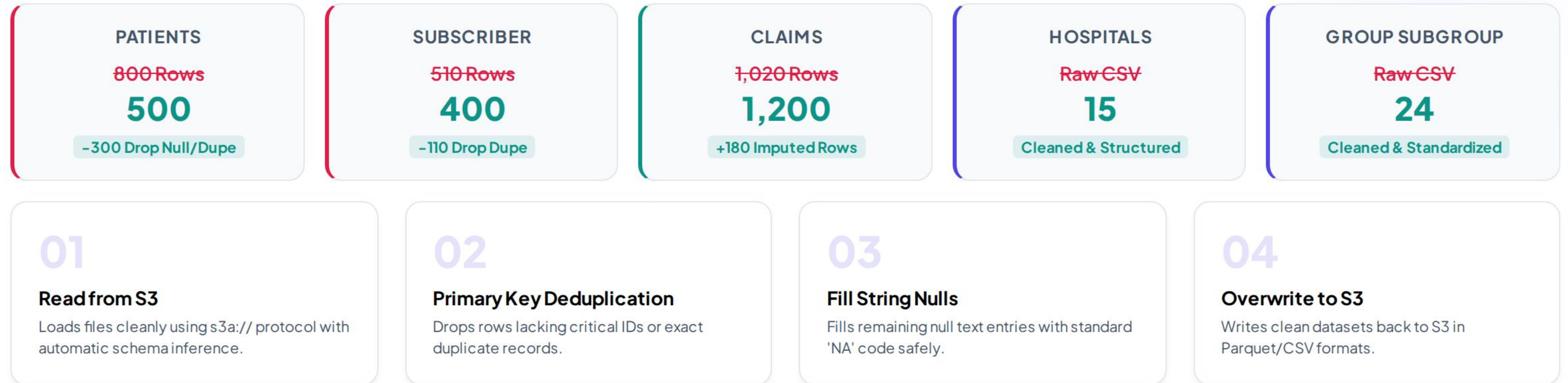
Resultant tables for the 13 use cases

 IAM Bucket Policies Confirmed Active

Data Cleaning: PySpark via PyCharm

05/14

We developed robust distributed PySpark cleaning scripts locally using PyCharm, optimizing performance on big data transformations before warehouse loading.



Warehouse Config: Amazon Redshift Serverless

06/14

PRODUCTION WORKGROUP DETAILS

takeo-project-workgroup

Leverages Amazon Redshift Serverless auto-scaling capabilities to reduce running compute costs during inactive query windows.

Namespace	capstone-project-namespace
AWS Region	us-east-2 (Ohio)
JDBC Port	5439 (Secured)

Active Endpoint

takeo-project-workgroup.187711699698.us-east-2.redshift-serverless.amazonaws.com

REDSHIFT RPUS

50

Balanced Capacity

AVG. EXEC TIME

< 1s

High-Performance OLAP

Data Load Validation: All Staging Tables Loaded

07/14

We performed verification row counts across all 5 staging schemas in Redshift Serverless using high-performance COPY loads.

Staging Table	Records Loaded	Data Status
staging.patients	500	✓ Cleaned
staging.subscriber	400	✓ Cleaned
staging.claims	1,200	✓ Cleaned
staging.hospitals	15	✓ Valid
staging.group_subgroup	24	✓ Valid

2,139

TOTAL STAGED ROWS

Full data validation passed with 0 truncated or skipped load errors during copy phase execution.

13 Core Analytical Business Use Cases Met

08/14

End-to-End Analytical Targets

UC01 Top Diseases Claim Counts

UC02 Subscribers Age Under 30

UC03 Group with Maximum Subgroups

UC04 Hospitals Serving Most Patients

UC05 Most Subscribed Subgroup

UC06 Total Rejected Claims Aggregation

UC07 Cities Generating Highest Claims

UC08 Govt vs Private Plan Preferences

UC09 Average Monthly Premium by State

UC10 Most Profitable Corporate Groups

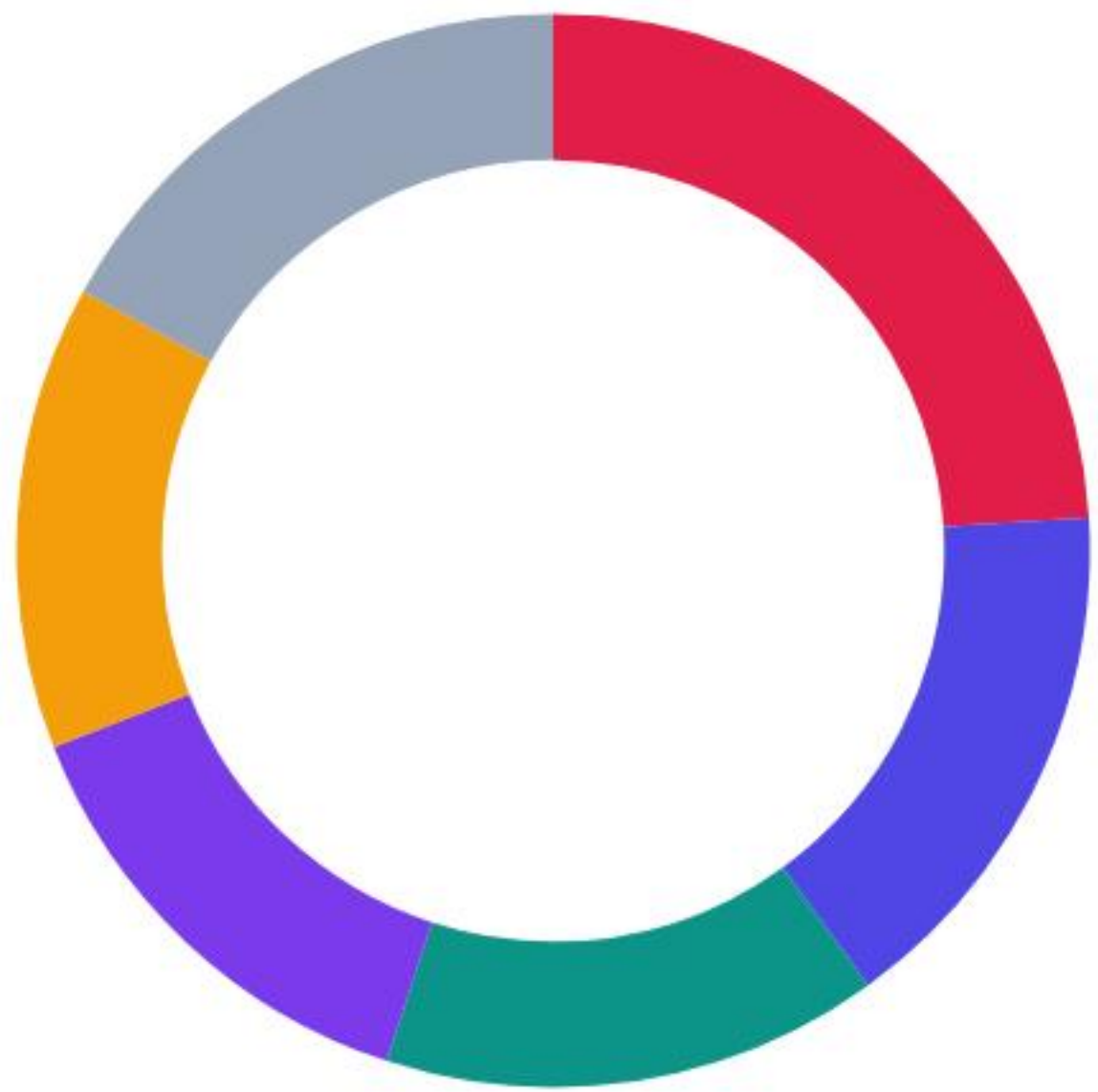
UC11 Cancer Patients Under 18 Counts

UC12 Cashless Claims Exceeding 50K

UC13 Females Age 40+ Having Knee Surgery Historical Claims Analysis

UC-01: Disease Claims Distribution Proportions

09/14



Cancer Leads with 212 Total Claims

Aggregated diagnostic data indicates Oncology claims represent nearly a quarter of all active policy pay-outs, showing an actionable target for preventative corporate wellness initiatives.

■ Oncology / Cancer (24%)

■ Renal Care (15%)

■ Orthopedics (14%)

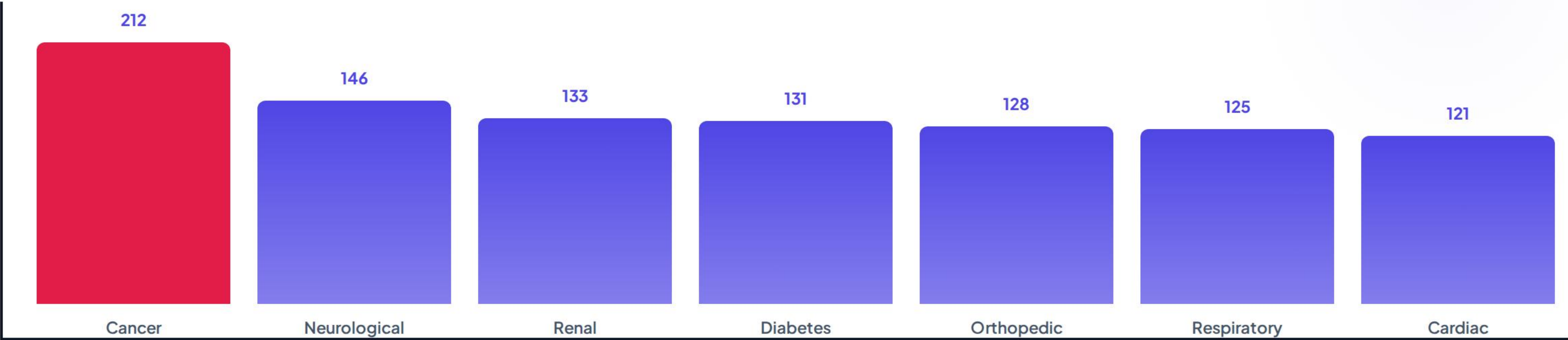
■ Neurological (16%)

■ Diabetes / Endocrine (14%)

■ Other Categories (17%)

Disease Category vs Claim Counts

Consolidated overview of total claim volumes generated by main health diagnosis categories within the insurance analytics cluster.



Scalable Database Schema Breakdown

11/14

staging schema

5 TABLES

Maintains cleaned baseline tables mirrored directly from our Silver layer in Amazon S3.

- patients (500 rows)
- subscriber (400 rows)
- claims (1200 rows)
- hospitals (15 rows)
- group_subgroup (24 rows)

project_output schema

13 TABLES

Consists of use case script result targets. Serves downstream analytical tools directly.

- uc01_top_disease_by_claims
- uc02_subscribers_under30
- uc03_group_max_subgroups
- uc04_hospital_most_patients
- ... up to uc13 script targets

 **Analytical Query Execution Speed:** Fast sub-second index metadata execution.

53ms Elapsed Time

Agile Delivery & Jira Project Tracking

12/14



13

Stories Completed



32

Story Points Delivered



100%

Sprint 2 Success Rate

Agile Pipeline Milestones

HC-01 Requirements Spec Document

DONE

HC-02 System Solution Design v1.0

DONE

HC-04 to HC-08 PySpark Core Cleanup

DONE

HC-11 to HC-13 Redshift Loading & Test

DONE

"Utilized a structured two-week agile sprint planning cycle to systematically track dependencies between raw ingestion, distributed data cleaning, schema optimization, and deployment validation stages."

Core Strategic Business Takeaways

13/14

Oncology Risk Insights

Oncology and cancer claims generate the highest overall claim financial volume (212 claims, representing over 24% of pay-out costs), highlighting a strategic area for specialized corporate preventative programs.

Mumbai Claims Hub

Geographic reporting indicates Mumbai as the highest volume region across all product categories, suggesting high-impact partnership opportunities with regional local hospitals.

Cashless Over Reimbursement

A significant majority of policyholders actively choose cashless claim settlement paths over traditional reimbursement, driving an operational mandate to digitize claims processing.

Corporate Plans Profitability

Corporate plan groups remain the primary monthly premium revenue driver, reinforcing the value of targeted business-to-business plan customization.

Project Submission Summary

14/14



Repository Live

Healthcare Insurance Capstone repository loaded with clean scripts, SQL use cases, and pipeline definitions.

[/healthcare-capstone-project](#)



SRS Document v1.0

Full Functional & Non-Functional Specifications mapped accurately to Sprint delivery workflows.



System Solution Design

Complete schema maps, DDL definitions, and optimized multi-layer storage architecture layout details.



Project Execution Evidence

Over 14 annotated test verification snapshots validating Redshift execution paths and PySpark code logic.